

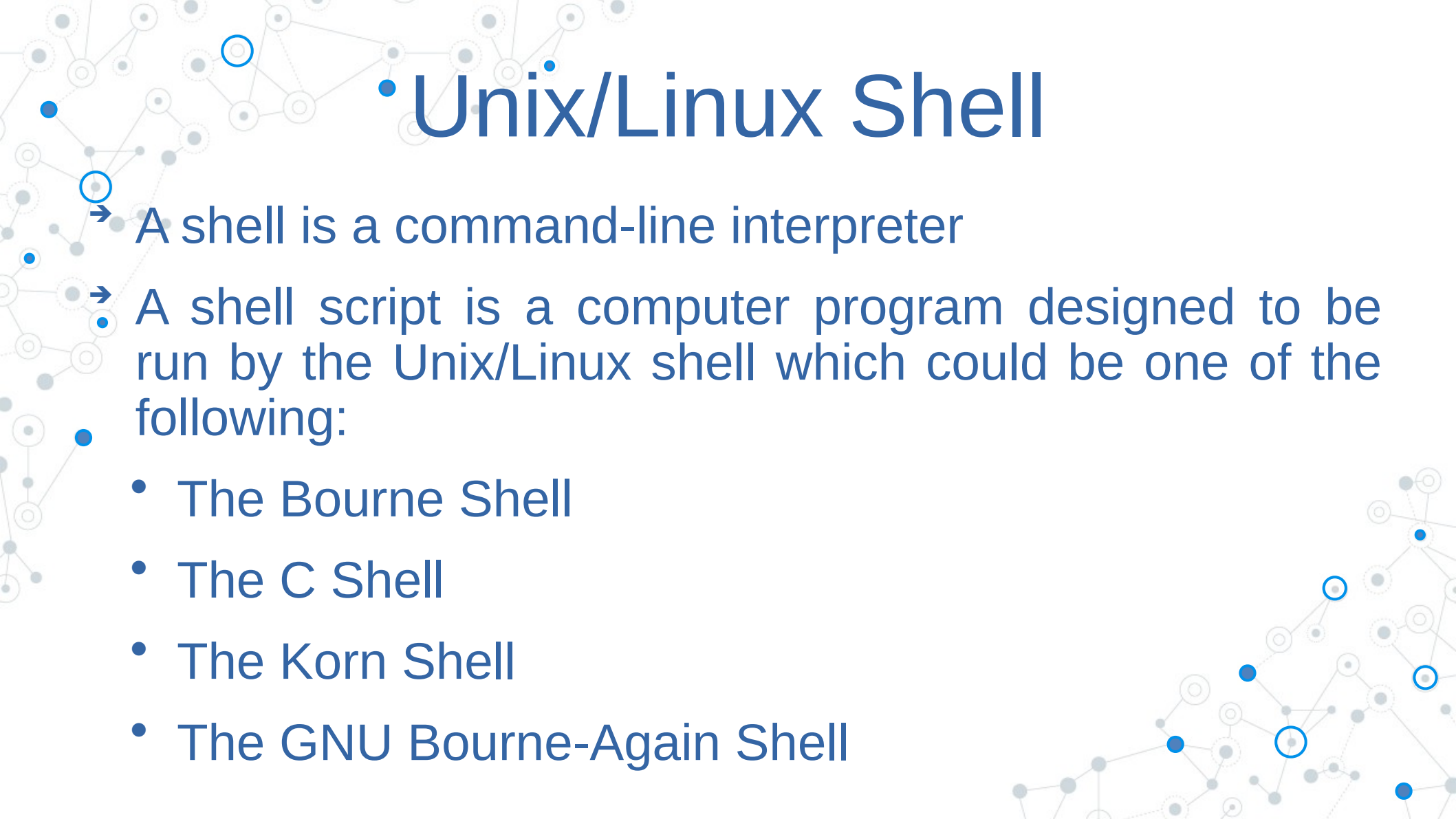


Shell Scripting

Operating System (CSC1036 & INF1036)



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• Unix/Linux Shell

- A shell is a command-line interpreter
- A shell script is a computer program designed to be run by the Unix/Linux shell which could be one of the following:
 - The Bourne Shell
 - The C Shell
 - The Korn Shell
 - The GNU Bourne-Again Shell

ls command

- The ls command without any options lists files and directories in a plain format without displaying much information like file types, permissions, modified date and time to mention just but a few.

```
jamie@debian:~$ ls
Battle.net-Setup.exe      metamorphose2_0.8.2-1_all.deb  Pictures
Desktop                  Music                          Public
Documents                Neofetch                      Templates
Downloads                pdfsam_3.3.6-1_all.deb        Videos
Linux_for_beginners.pdf  PDFsam_merge.pdf
```

ls -l output

\$ls -l

drwxrwxr-x 2 amrood amrood 4096 Dec 25 09:59 uml

-rw-rw-r-- 1 amrood amrood 5341 Dec 25 08:38 uml.jpg

drwxr-xr-x 2 amrood amrood 4096 Feb 15 2006 univ

drwxr-xr-x 2 root root 4096 Dec 9 2007 urlspedia

-rw-r--r-- 1 root root 276480 Dec 9 2007 urlspedia.tar

drwxr-xr-x 8 root root 4096 Nov 25 2007 usr

drwxr-xr-x 2 200 300 4096 Nov 25 2007 webthumb-1.01

-rwxr-xr-x 1 root root 3192 Nov 25 2007 webthumb.php

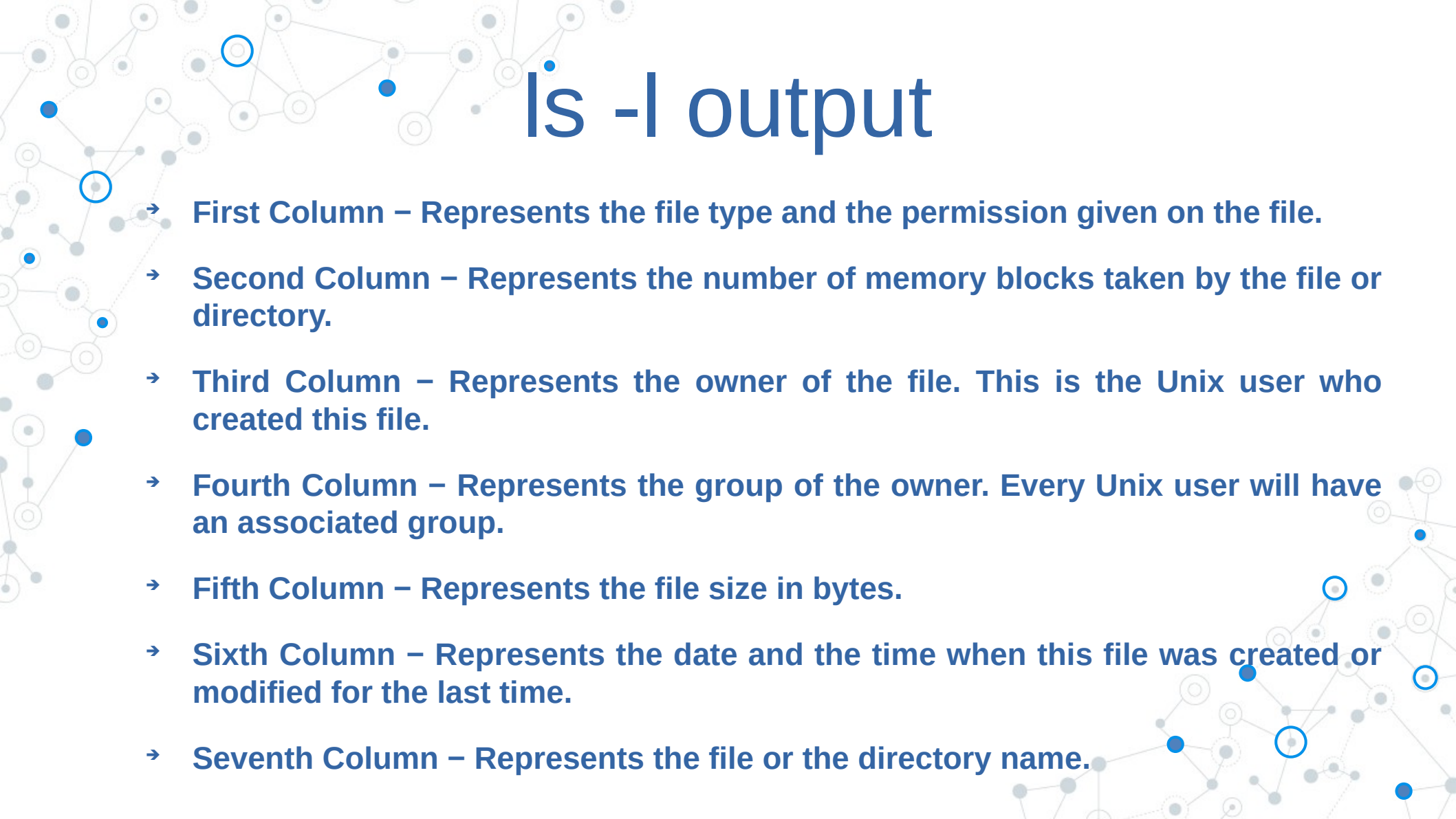
-rw-rw-r-- 1 amrood amrood 20480 Nov 25 2007 webthumb.tar

-rw-rw-r-- 1 amrood amrood 5654 Aug 9 2007 yourfile.mid

-rw-rw-r-- 1 amrood amrood 166255 Aug 9 2007 yourfile.swf

drwxr-xr-x 11 amrood amrood 4096 May 29 2007 zlib-1.2.3

\$

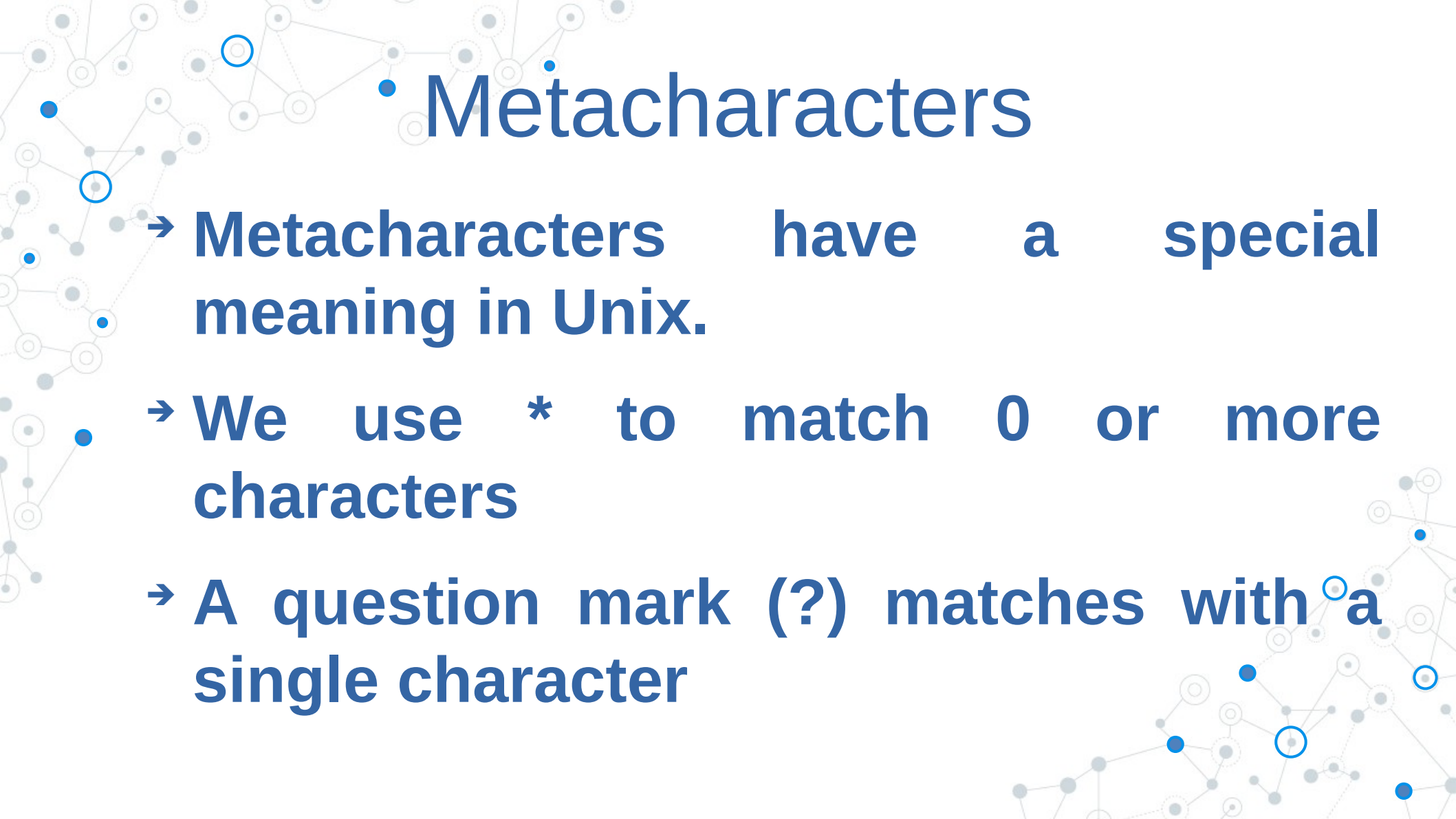


ls -l output

- **First Column** – Represents the file type and the permission given on the file.
- **Second Column** – Represents the number of memory blocks taken by the file or directory.
- **Third Column** – Represents the owner of the file. This is the Unix user who created this file.
- **Fourth Column** – Represents the group of the owner. Every Unix user will have an associated group.
- **Fifth Column** – Represents the file size in bytes.
- **Sixth Column** – Represents the date and the time when this file was created or modified for the last time.
- **Seventh Column** – Represents the file or the directory name.

ls -l output

—	Regular file, such as an ASCII text file, binary executable, or hard link.
b	Block special file. Block input/output device file such as a physical hard drive.
c	Character special file. Raw input/output device file such as a physical hard drive.
d	Directory file that contains a listing of other files and directories.
l	Symbolic link file. Links on any regular file.
p	Named pipe. A mechanism for interprocess communications.
s	Socket used for interprocess communication.



• Metacharacters

- Metacharacters have a special meaning in Unix.
- We use * to match 0 or more characters
- A question mark (?) matches with a single character



• Metacharacters

→ **\$ls ch*.doc**

→ **ch01-1.doc**
ch03-2.doc
ch05.doc
ch02-1.doc

ch010.doc
ch04-1.doc
ch06-2.doc

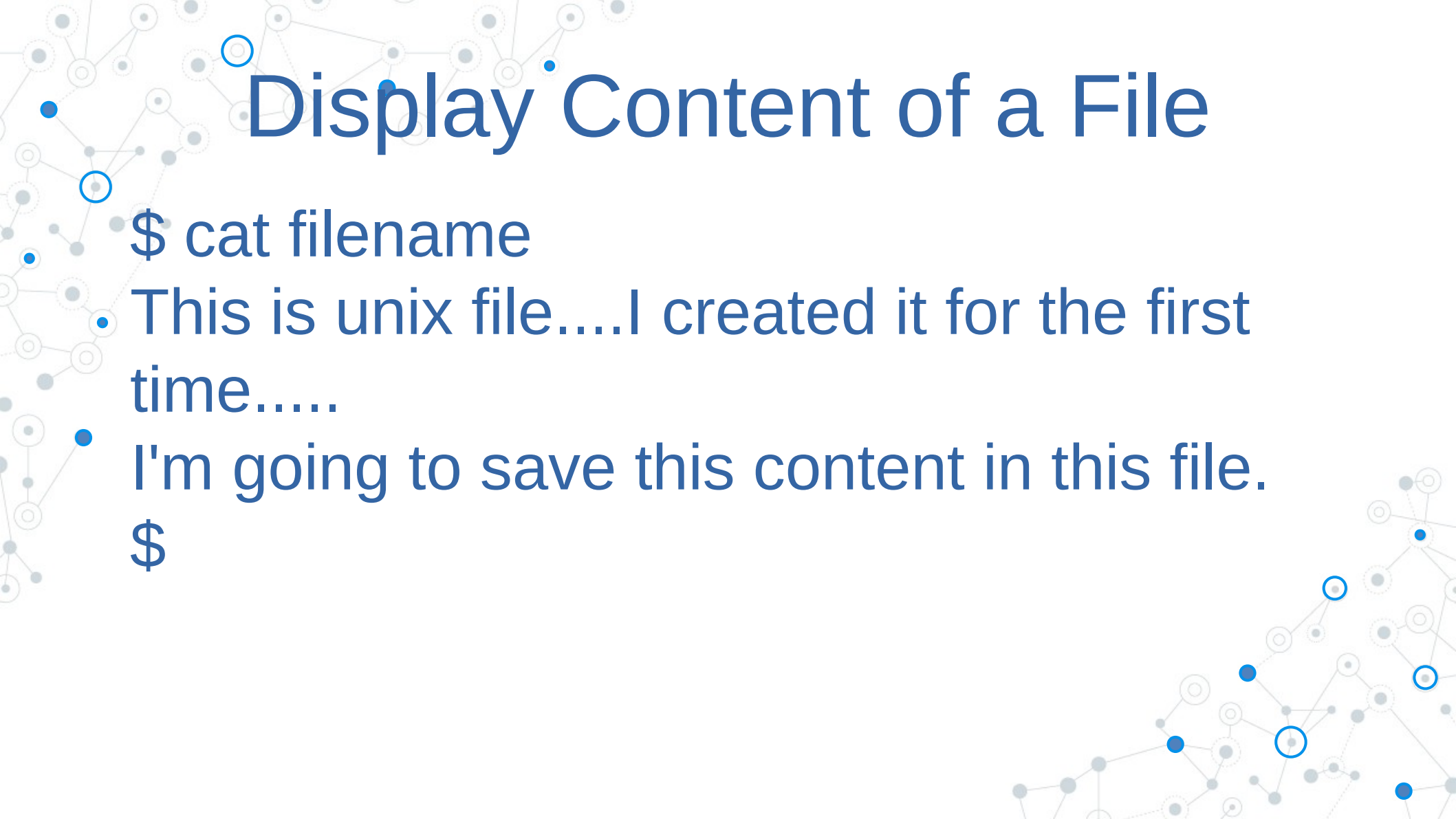
ch02.doc
ch040.doc
ch01-2.doc



Hidden Files

```
$ ls -a
```

```
.      .profile      docs      lib      test_results  
..     .rhosts       hosts     pub      users  
.emacs  bin          hw1       res.01   work  
.exrc   ch07         hw2       res.02  
.kshrc  ch07.bak     hw3       res.03  
$
```



Display Content of a File

\$ cat filename

• This is unix file....I created it for the first time.....

• I'm going to save this content in this file.

\$

Line numbers through the -b option

```
$ cat -b filename
```

```
1 This is unix file....I created it for the first  
time.....
```

```
2 I'm going to save this content in this file.
```

```
$
```

Counting Words in a File

```
$ wc filename
```

```
2 19 103 filename
```

```
$
```

- First Column – Total number of lines in the file.
- Second Column – Total number of words in the file.
- Third Column – Total number of bytes in the file.
- Fourth Column – File name.



• Counting Words in Multiple Files

• `$ wc filename1 filename2 filename3`





Copying Files/Folders

\$ cp source-file destination-file

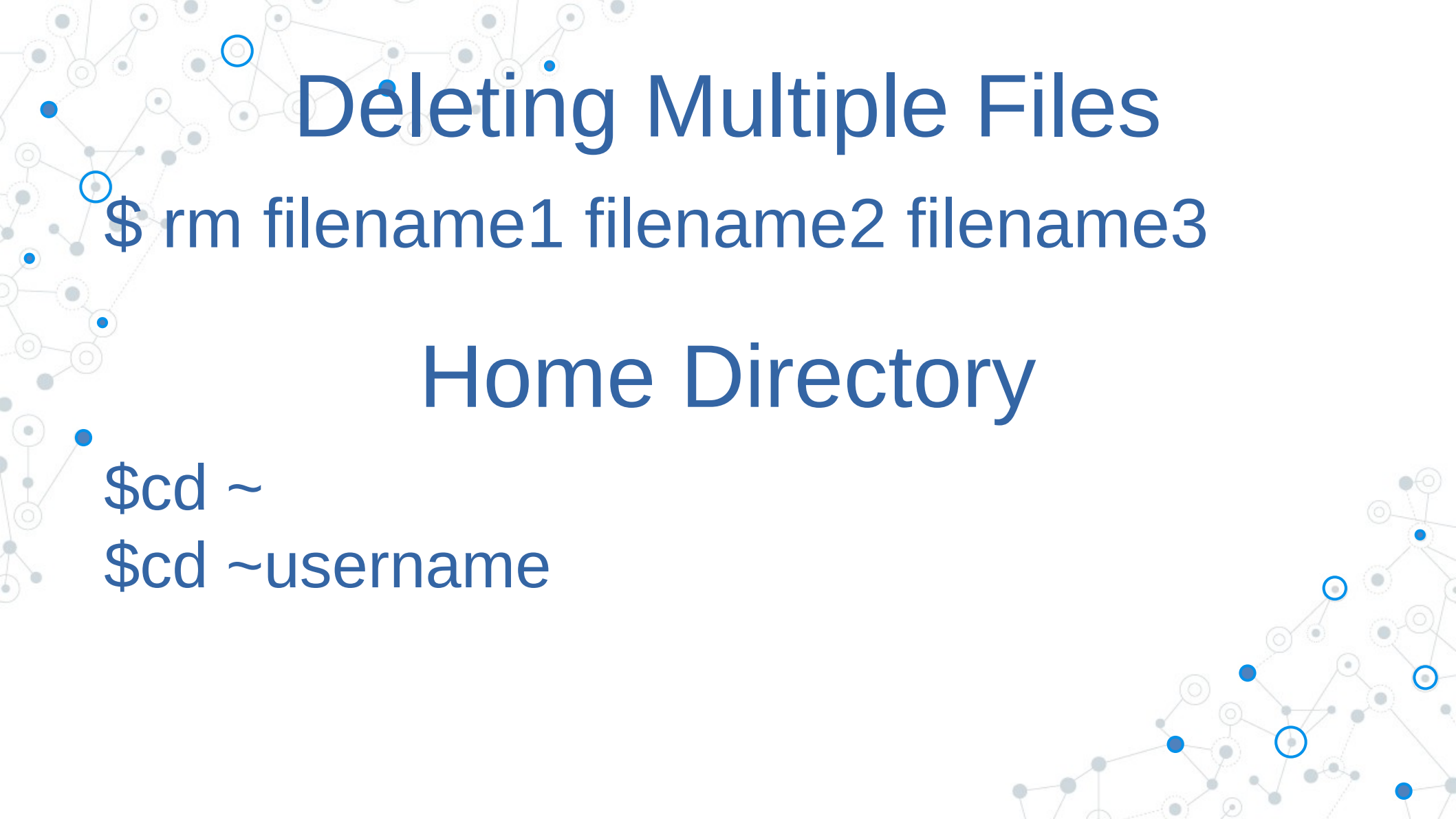
\$ cp -r source-folder destination-folder

Renaming Files

\$ mv old_file new_file

Deleting Files

\$ rm filename



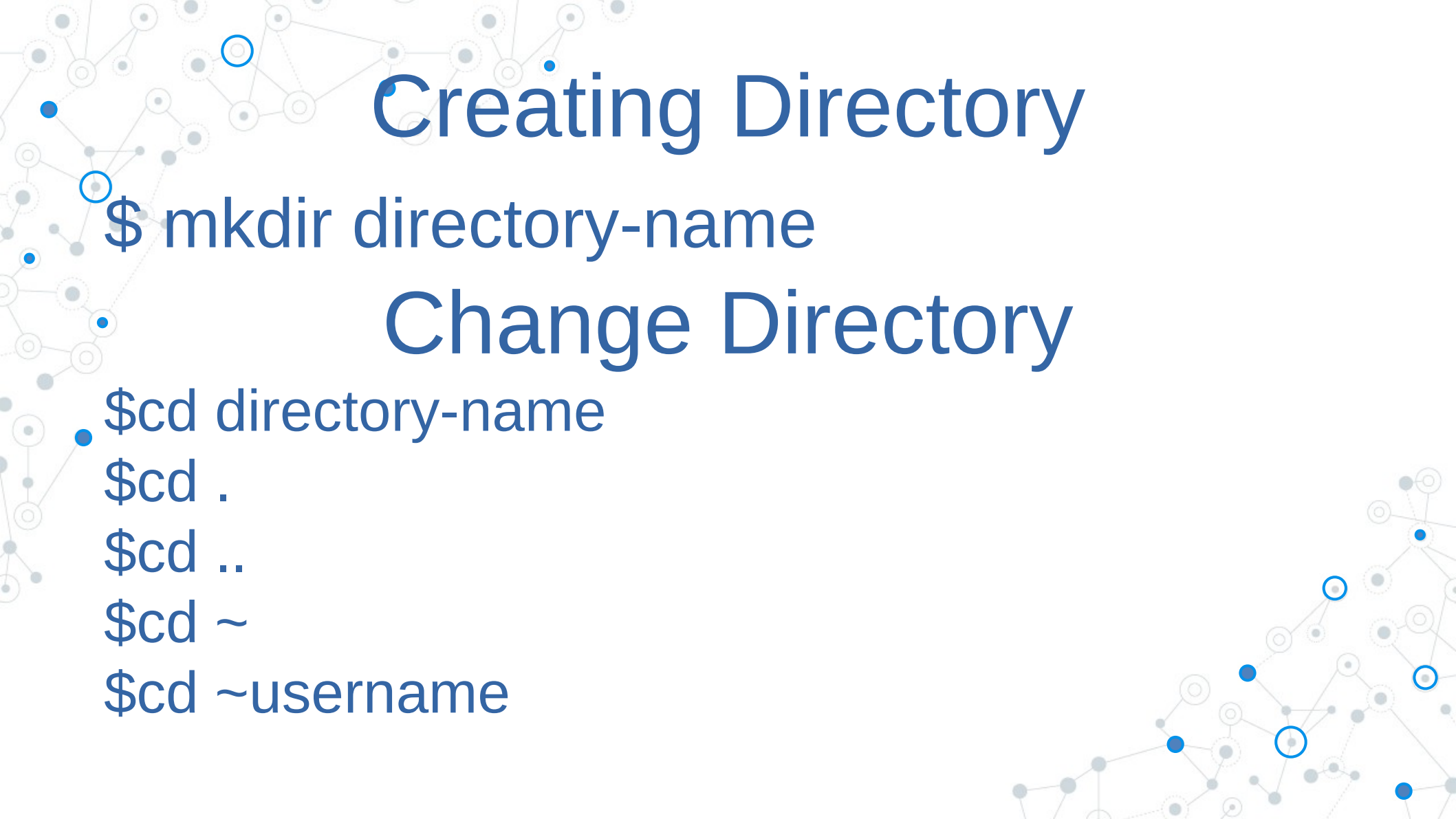
Deleting Multiple Files

```
$ rm filename1 filename2 filename3
```

Home Directory

```
$ cd ~
```

```
$ cd ~username
```



Creating Directory

```
$ mkdir directory-name
```

Change Directory

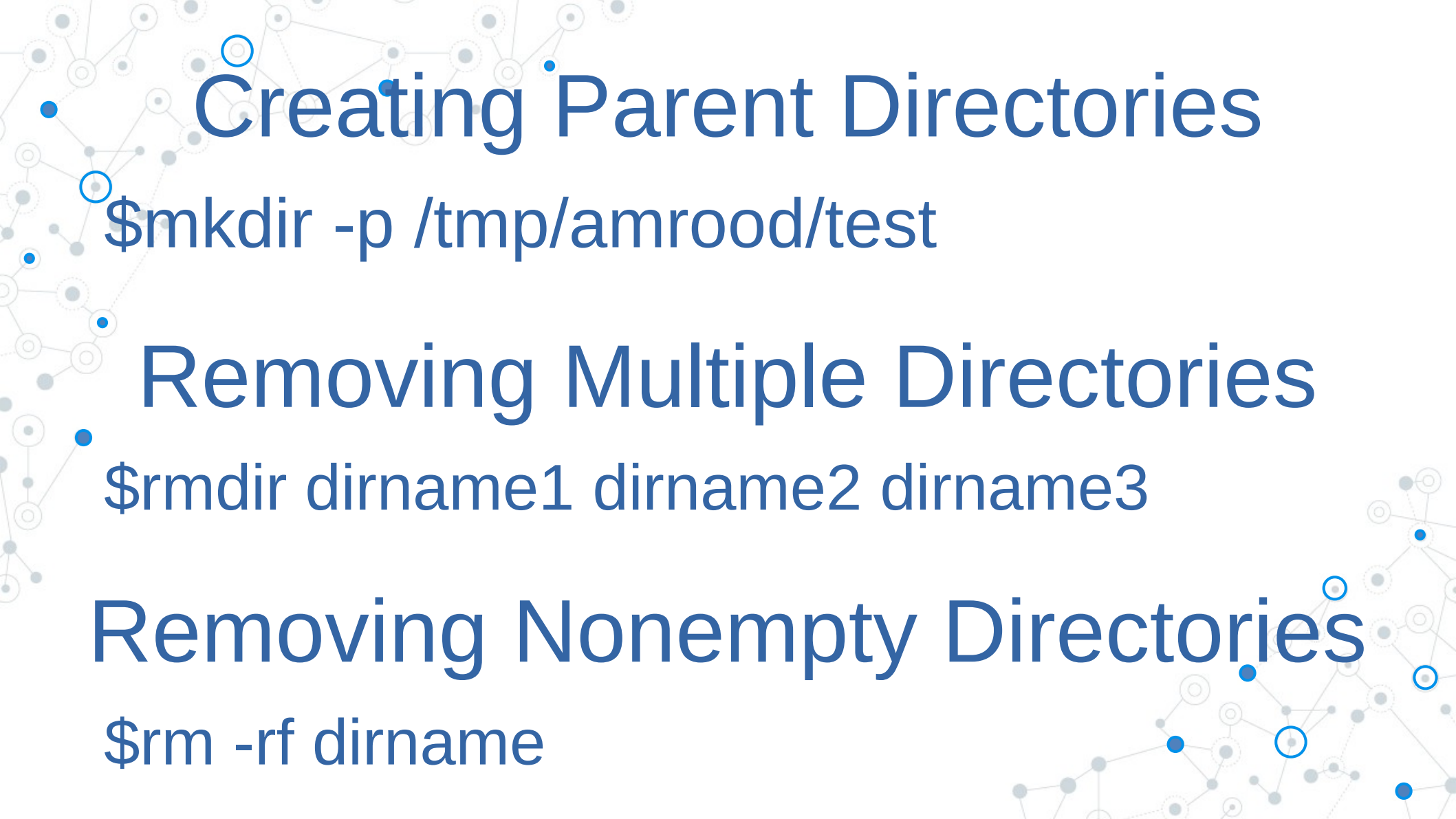
```
$cd directory-name
```

```
$cd .
```

```
$cd ..
```

```
$cd ~
```

```
$cd ~username
```

Creating Parent Directories

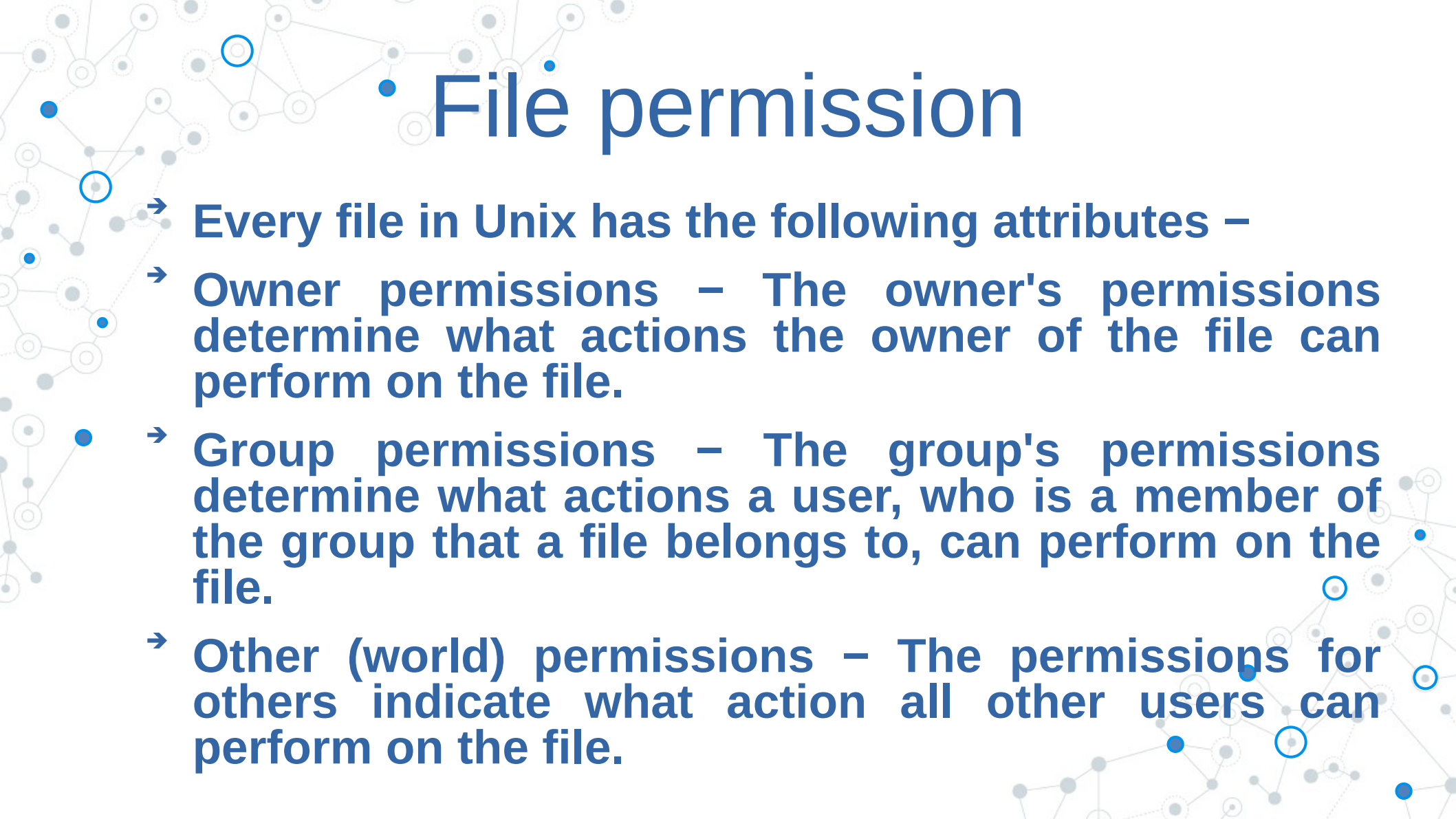
```
$mkdir -p /tmp/amrood/test
```

Removing Multiple Directories

```
$rmdir dirname1 dirname2 dirname3
```

Removing Nonempty Directories

```
$rm -rf dirname
```

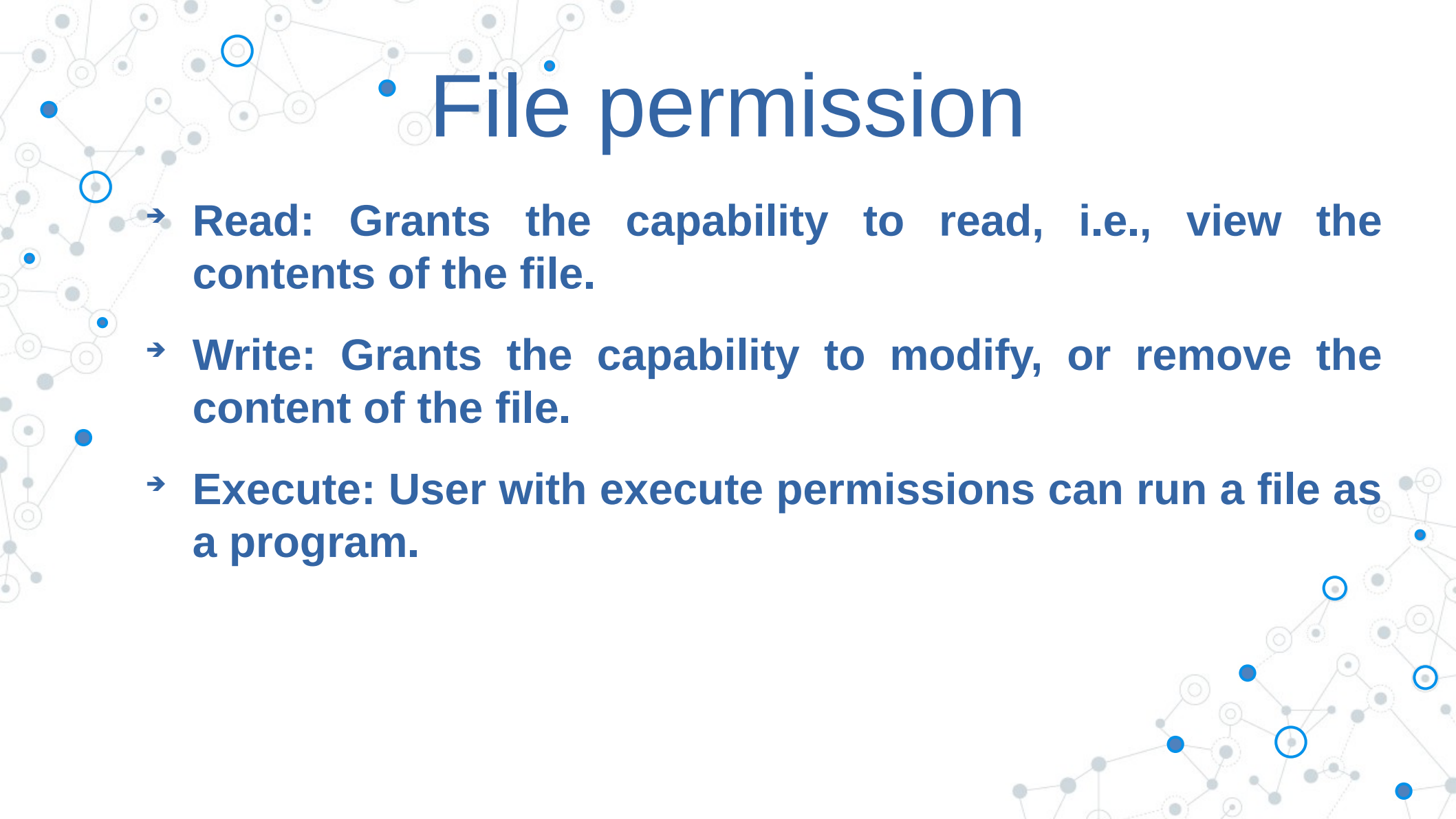


• File permission

- Every file in Unix has the following attributes –
- Owner permissions – The owner's permissions determine what actions the owner of the file can perform on the file.
- Group permissions – The group's permissions determine what actions a user, who is a member of the group that a file belongs to, can perform on the file.
- Other (world) permissions – The permissions for others indicate what action all other users can perform on the file.

• File permission

- `$ls -l /home/amrood`
- `-rwxr-xr-- 1 amrood users 1024 Nov 2 00:10 myfile`
- `drwxr-xr--- 1 amrood users 1024 Nov 2 00:10 mydir`
- The first three characters (2-4) represent the permissions for the file's owner.
- The second group of three characters (5-7) consists of the permissions for the group to which the file belongs.
- The last group of three characters (8-10) represents the permissions for everyone else.



• File permission

- **Read:** Grants the capability to read, i.e., view the contents of the file.
- **Write:** Grants the capability to modify, or remove the content of the file.
- **Execute:** User with execute permissions can run a file as a program.

Directory Access Modes

- **Read:** Access to a directory means that the user can read the contents. The user can look at the filenames inside the directory.
- **Write:** Access means that the user can add or delete files from the directory.
- **Execute:** Executing a directory doesn't really make sense, so think of this as a traverse permission. Execute permission allows you to enter the directory (i.e., `cd` into it), and to access any of its files.

Changing Permission

```
$ls -l testfile
```

```
-rwxrwxr-- 1 amrood  users 1024 Nov 2 00:10 testfile
```

```
$chmod o+wx testfile
```

```
$ls -l testfile
```

```
-rwxrwxrwx 1 amrood  users 1024 Nov 2 00:10 testfile
```

```
$chmod u-x testfile
```

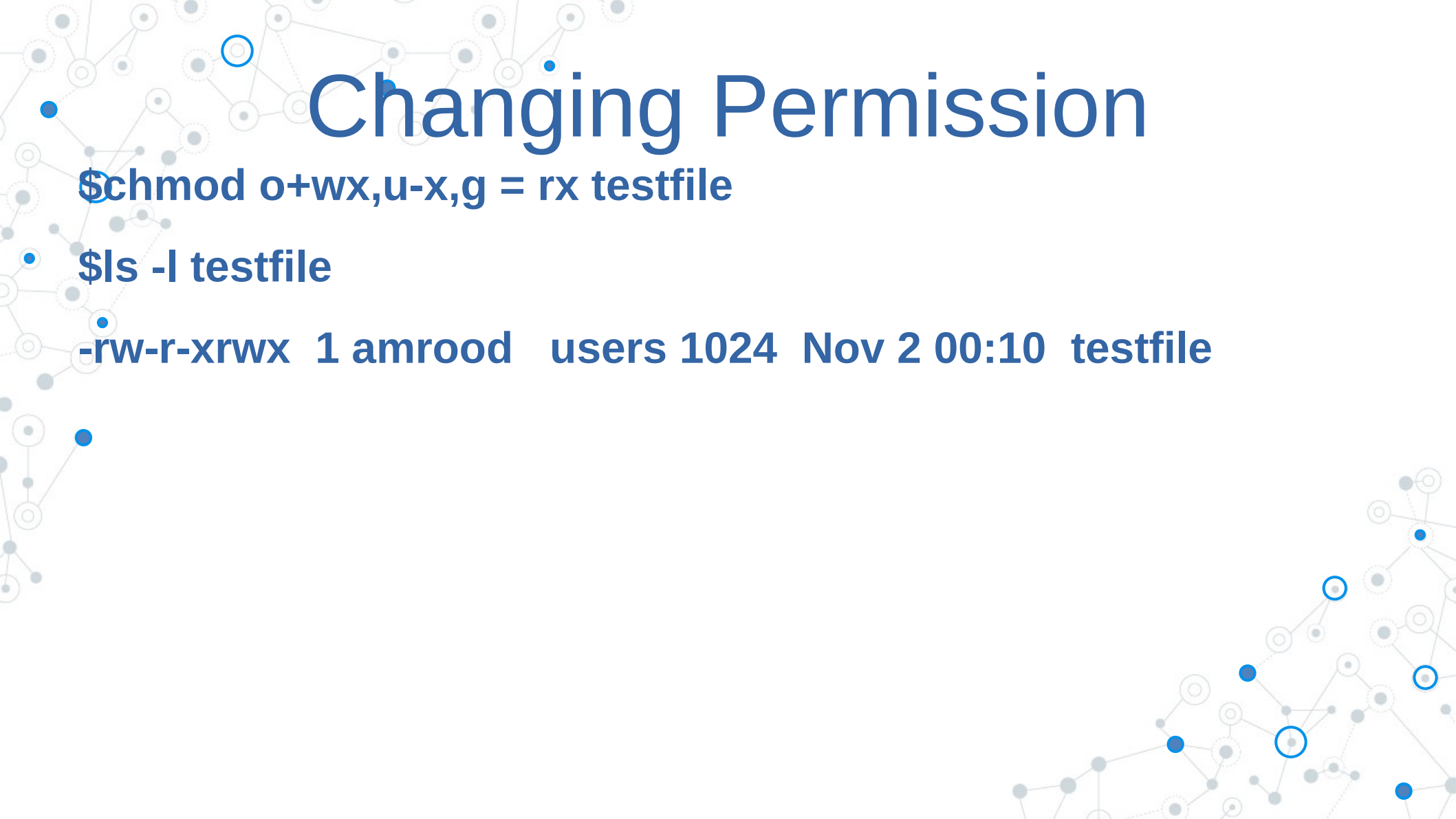
```
$ls -l testfile
```

```
-rw-rwxrwx 1 amrood  users 1024 Nov 2 00:10 testfile
```

```
$chmod g = rx testfile
```

```
$ls -l testfile
```

```
-rw-r-xrwx 1 amrood  users 1024 Nov 2 00:10 testfile
```



Changing Permission

`$chmod o+wx,u-x,g = rx testfile`

`$ls -l testfile`

`-rw-r-xrwx 1 amrood users 1024 Nov 2 00:10 testfile`

Changing Permission

Number	Octal Permission Representation	Ref
0	No permission	---
1	Execute permission	--x
2	Write permission	-w-
3	Execute and write permission: 1 (execute) + 2 (write) = 3	-wx
4	Read permission	r--
5	Read and execute permission: 4 (read) + 1 (execute) = 5	r-x
6	Read and write permission: 4 (read) + 2 (write) = 6	rw-
7	All permissions: 4 (read) + 2 (write) + 1 (execute) = 7	rwX

Changing Permission

```
$ls -l testfile
```

```
-rwxrwxr-- 1 amrood  users 1024 Nov 2 00:10 testfile
```

```
$ chmod 755 testfile
```

```
$ls -l testfile
```

```
-rwxr-xr-x 1 amrood  users 1024 Nov 2 00:10 testfile
```

```
$chmod 743 testfile
```

```
$ls -l testfile
```

```
-rwxr---wx 1 amrood  users 1024 Nov 2 00:10 testfile
```

```
$chmod 043 testfile
```

```
$ls -l testfile
```

```
----r---wx 1 amrood  users 1024 Nov 2 00:10 testfile
```

• ENVIRONMENTAL VARIABLES

➔ `$TEST="Unix Programming"`

➔ `$echo $TEST`

➔ When you log in to the system, the shell undergoes a phase called initialization to set up the environment. This is usually a two-step process that involves the shell reading the following files:

➔ `/etc/profile`

➔ `profile`

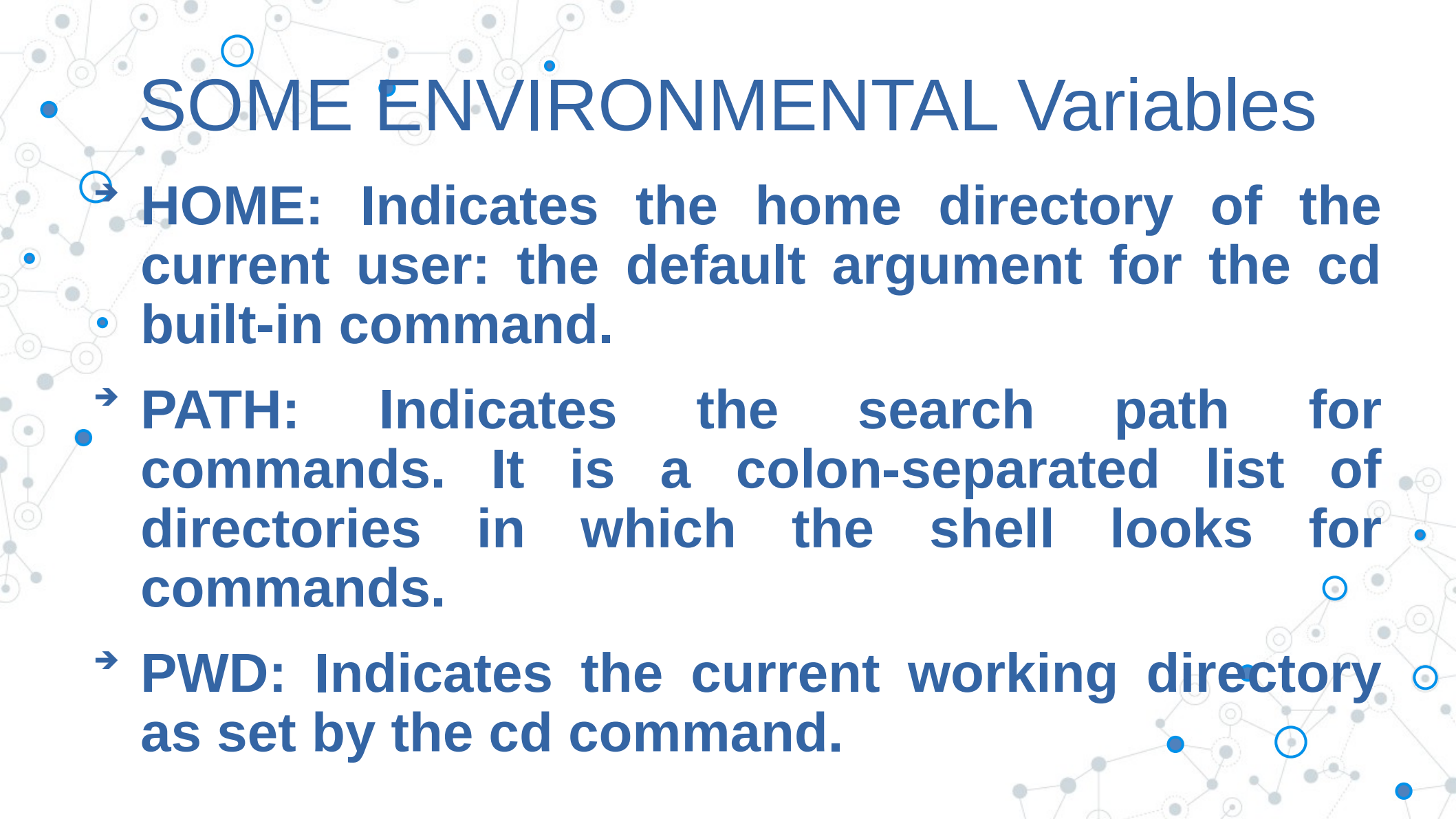
• ENVIRONMENTAL VARIABLES

→ The .profile File

- The file `/etc/profile` is maintained by the system administrator of your Unix machine and contains shell initialization information required by all users on a system.
- The file `~/.profile` is under your control.
- `~/.bash_profile`, `~/.bashrc`, or `~/.profile`

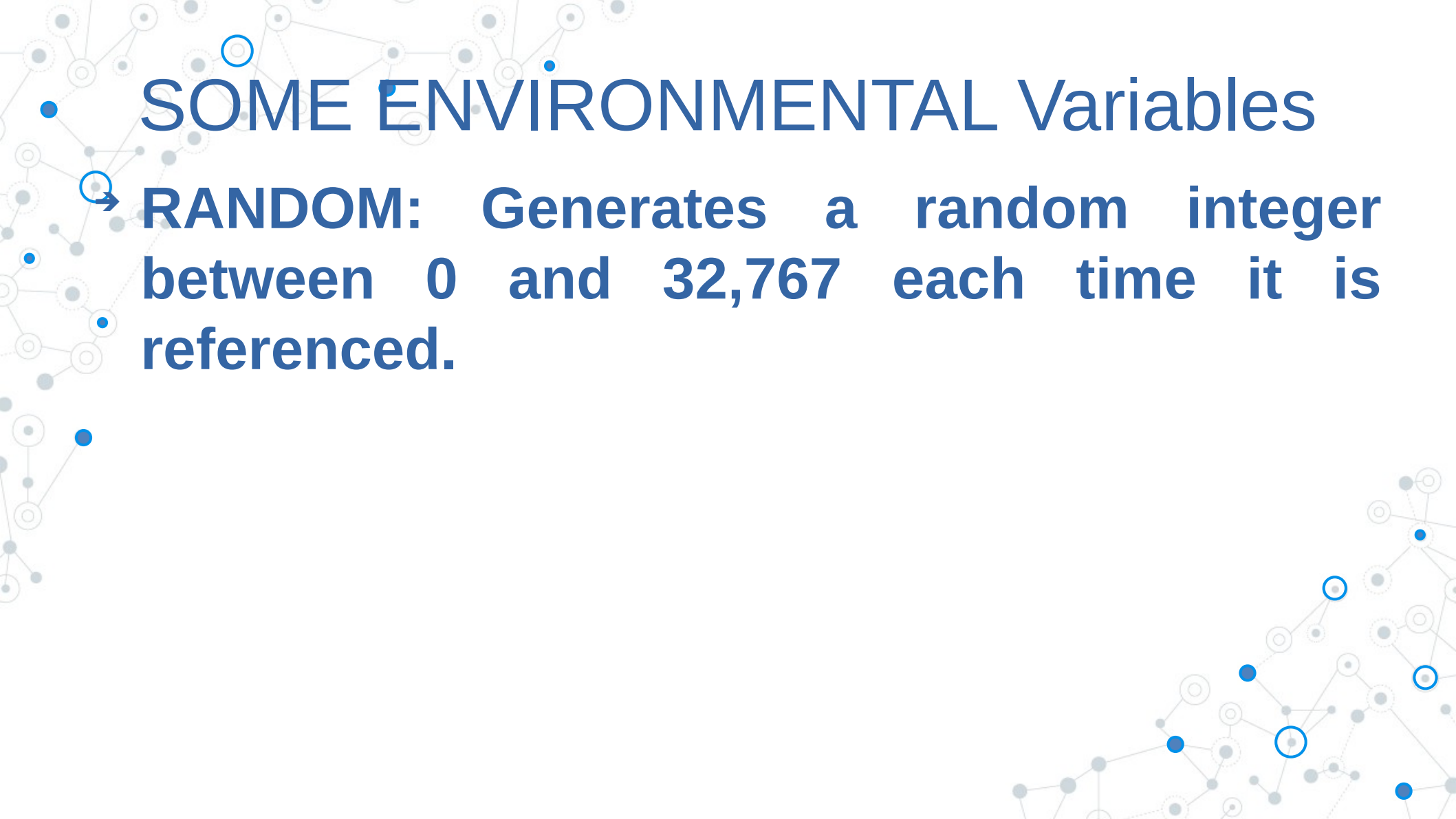
PATH

- The PATH variable specifies the locations in which the shell should look for commands.
- \$hello
- hello: not found
- echo \$PATH
- export PATH=\$PATH:/place/with/the/file



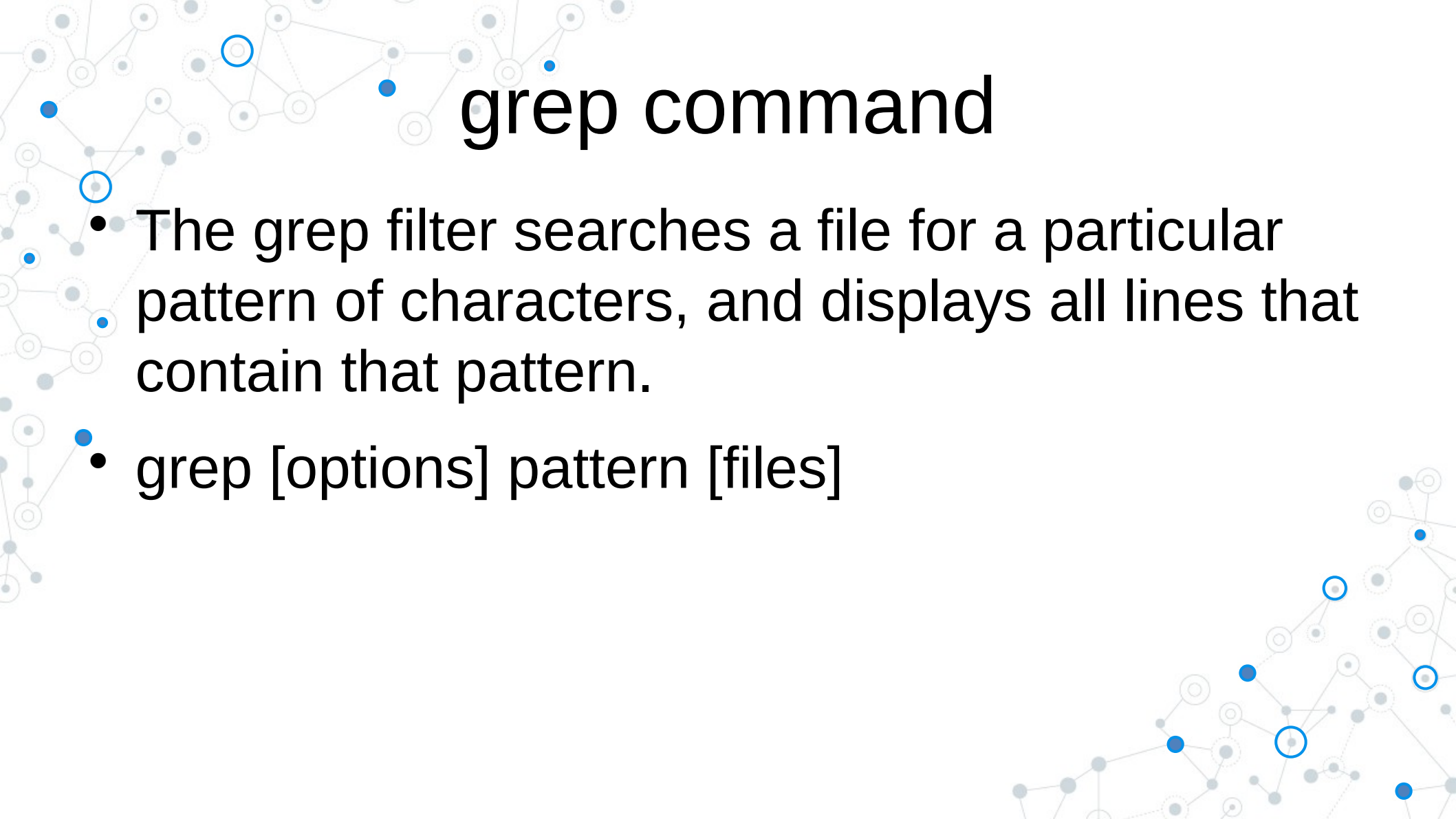
• SOME ENVIRONMENTAL Variables

- ➔ **HOME:** Indicates the home directory of the current user: the default argument for the cd built-in command.
- ➔ **PATH:** Indicates the search path for commands. It is a colon-separated list of directories in which the shell looks for commands.
- ➔ **PWD:** Indicates the current working directory as set by the cd command.



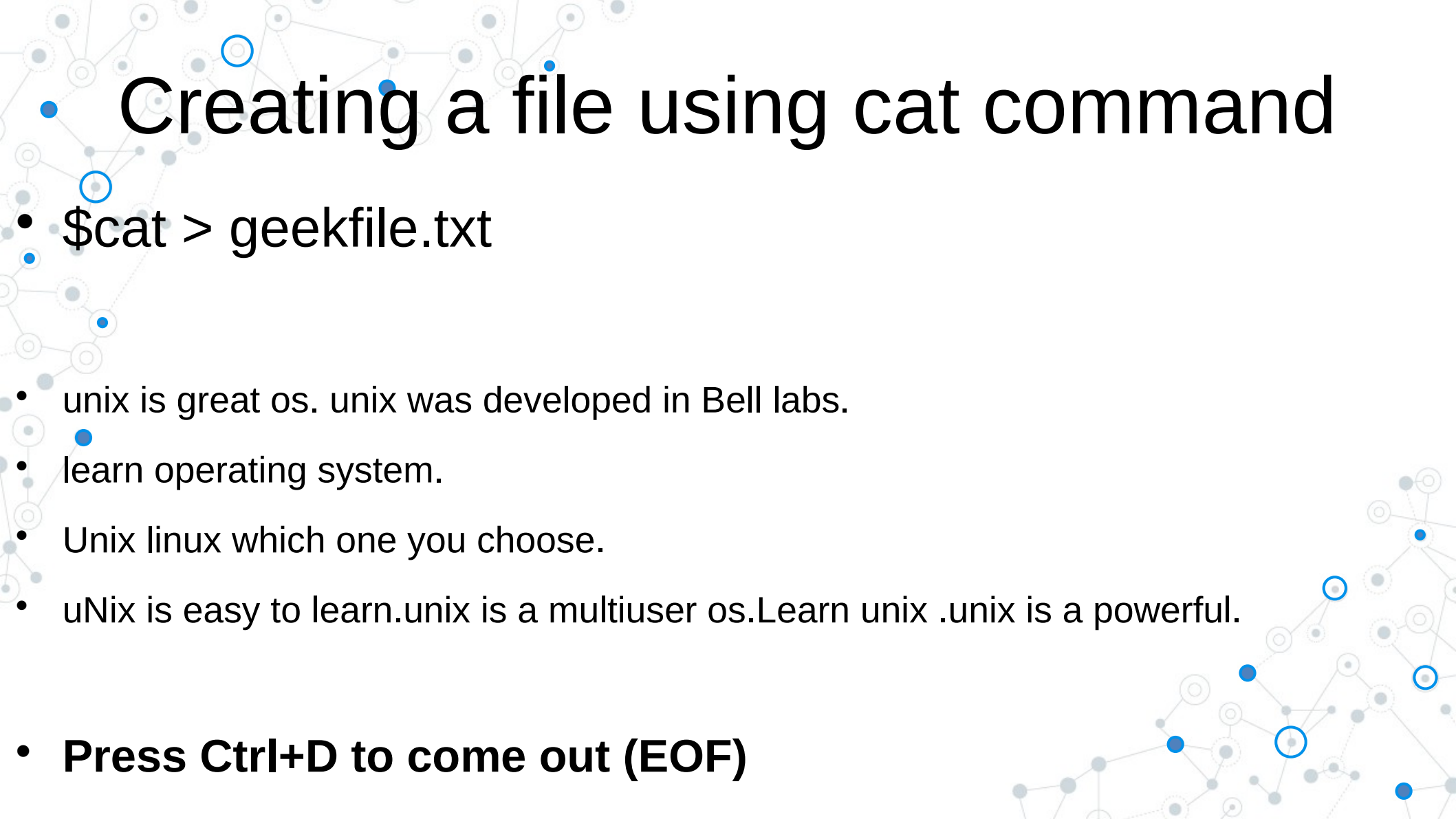
• SOME ENVIRONMENTAL Variables

→ **RANDOM:** Generates a random integer between 0 and 32,767 each time it is referenced.



grep command

- The grep filter searches a file for a particular pattern of characters, and displays all lines that contain that pattern.
- `grep [options] pattern [files]`



• Creating a file using cat command

- `$cat > geekfile.txt`
- unix is great os. unix was developed in Bell labs.
- learn operating system.
- Unix linux which one you choose.
- uNix is easy to learn.unix is a multiuser os.Learn unix .unix is a powerful.
- **Press Ctrl+D to come out (EOF)**

Unix/Linux Commands

→ Touch: used to create a new empty file and to change the timestamps of existing files

touch filename

→ Ln: This command is used to create hard or soft links

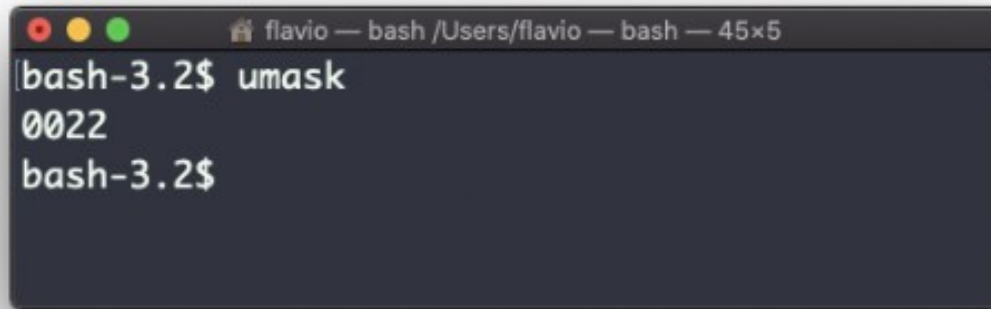
- Hard: ln <original> <link> (Indistinguishable)
- Soft: ln -s <original> <link>

Unix/Linux Commands

- Alias: Used to display defined aliases and to create a new alias
 - `alias hh="ls -l"`
- Umask: Default Permissions can be controlled and modified using the umask command
 - Typing umask with no arguments will show you the current umask

Unix/Linux Commands

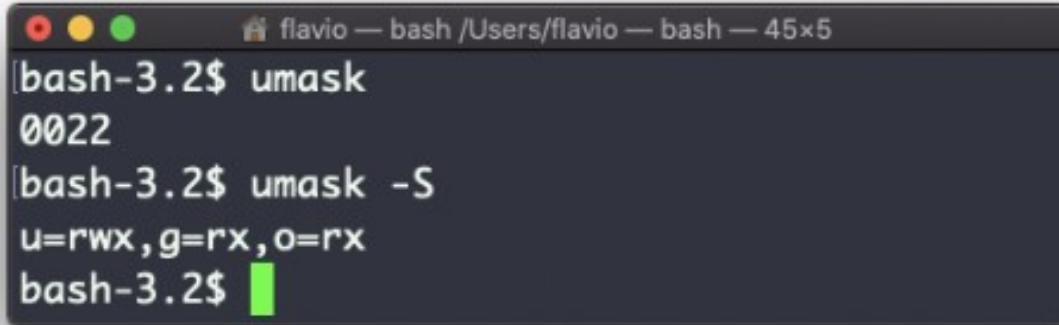
- Typing `umask` with no arguments will show you the current umask

A screenshot of a terminal window with a dark background. The window title bar shows 'flavio — bash /Users/flavio — bash — 45x5'. The terminal text shows the command 'umask' being entered at the 'bash-3.2\$' prompt, followed by the output '0022' on the next line, and the prompt 'bash-3.2\$' on the third line.

```
flavio — bash /Users/flavio — bash — 45x5
bash-3.2$ umask
0022
bash-3.2$
```

- Use `umask -S` to see a human-readable notation:

Unix/Linux Commands

A terminal window with a dark background and light text. The title bar shows 'flavio — bash /Users/flavio — bash — 45x5'. The terminal content shows the execution of 'umask' commands: first 'umask' returns '0022', then 'umask -S' returns 'u=rwx,g=rx,o=rx'. The prompt 'bash-3.2\$' is followed by a green cursor bar.

```
flavio — bash /Users/flavio — bash — 45x5
bash-3.2$ umask
0022
bash-3.2$ umask -S
u=rwx,g=rx,o=rx
bash-3.2$
```

- We can set a new value for the mask
- `umask 002`
- `umask g+r`

Unix/Linux Commands

- Clear: To clear the console window
- History: To display history of commands
- Export: The export command is used to export variables so that child processes can access it
 - A=5 vs export A=5
 - Shell scripts will not have access in case 1



Unix/Linux Commands

→ To remove an Exported Variable Test:

- `export -n TEST`

Redirection

- Output of a command can be written to a file using > operator
 - `echo "hello" > output.txt`
 - `echo "hello" >> output.txt` (Appends)

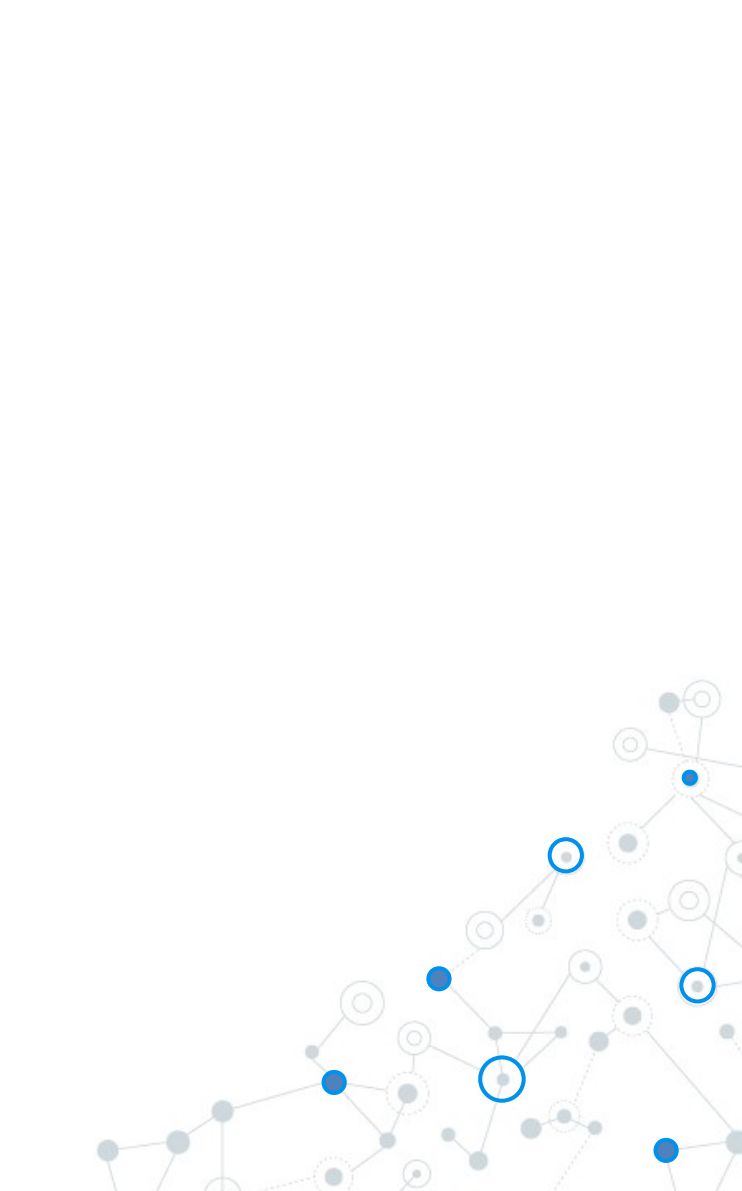
Sample Shell Script

- `#This is sample shell script`
- `echo "What is your name?"`
- `read PERSON`
- `echo "Hello, $PERSON"`

- `$/test.sh`
- `What is your name?`
- `Utpal`
- `Hello, Utpal`
- `$`

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with blue dots. The lines are thin and gray, creating a subtle background pattern.

Thank you

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a network of nodes and lines, with some nodes highlighted by blue circles and others by blue dots. The overall style is clean and modern, with a light gray background.